

AI VIDEO GENERATOR — V4 ANIMATION UPGRADE TASK

Based on the supplied explainer-video (22) result — move from “good presentation” to “active visual storytelling”.

Current result assessment: The new video is a clear improvement: it has a coherent visual style, characters, environments and synchronized narration. However, the frame-by-frame review still shows long periods where the characters are standing almost still while the narration describes major events. The next upgrade is therefore action density + visual event coverage, not simply prettier graphics.

Observed example: during the village/monster narration, the screen can remain on a mostly static village with Eira and Gorthok while the narration describes monsters emerging, villagers reacting and an attack. The next version must visibly stage those events as they are spoken.

1. THE NEW TARGET

OLD:
Narration → scene illustration → character stands → text/caption

V4:
Narration / dialogue
↓
Story beat
↓
Shot plan
↓
Character actions + object actions + environment actions
↓
Camera movement + staging
↓
Animation events
↓
Audio-timed render

Every important sentence should cause something meaningful to happen on screen.

2. ACTION-DENSITY REQUIREMENT

- Do not allow a story shot to remain visually static for more than ~2 seconds unless the script intentionally calls for a pause, emotional reaction or establishing shot.
- Every 3–6 seconds, introduce at least one meaningful visual change: character movement, camera move, expression change, object movement, environment movement, reveal, interaction, impact, transition or new composition.
- Do not count caption appearance as animation.
- Do not count a simple fade-in/fade-out as meaningful action.
- Do not satisfy action requirements by moving the entire scene as one flat image.
- Action must be semantically connected to the narration/dialogue.

3. STORY MODE — SHOT GRAMMAR

Each shot must contain:

```
SHOT
├─ camera
│   └─ framing
│   └─ position
```

```

|   | movement
|   | focus target
|---| environment
|   | background motion
|   | lighting
|   | atmosphere
|---| characters[]
|   | start pose
|   | action
|   | path
|   | expression
|   | gaze
|   | end pose
|---| props[]
|   | state
|   | interaction
|   | movement
|---| visualEvents[]
|---| narration
|---| dialogue
|---| timing

```

4. REQUIRED ACTION VOCABULARY

- Locomotion: idle, walk, run, approach, retreat, turn, crouch, jump, stumble.
- Gesture: point, wave, reach, grab, push, pull, lift, throw, hold, hand-over-object.
- Head/face: look_at, look_away, nod, shake_head, surprise, fear, anger, relief, smile.
- Interaction: open, close, inspect, repair, activate, deactivate, carry, place, pick_up, drop.
- Dialogue: speaking animation, listener reaction, turn-taking, gaze toward speaker.
- Environment: wind, smoke, dust, rain, fire/light, moving clouds, crowd movement, machine motion.
- Action effects: impact, shake, sparks, particles, debris, glow, screen flash, object deformation where appropriate.

5. ACTION-TO-NARRATION MAPPING

Example narration:

"The monsters emerged from the forest."

REQUIRED visual timeline:

```

0.0s  Camera faces forest.
0.8s  Leaves/branches move.
1.4s  First monster silhouette appears.
2.0s  Monster steps out.
2.8s  Second monster follows.
3.5s  Camera pulls wider.
4.0s  Villager notices movement.
4.5s  Villager turns toward forest.
5.0s  Expression changes to fear.
5.5s  Villager signals others.

```

The narration and visuals must describe the SAME event,
not two unrelated versions of the story.

6. CAMERA DIRECTOR — ADD CINEMATIC MOVEMENT

- Use establishing wide shots for locations.
- Use medium shots for character interaction.
- Use close-ups for emotion and important objects.
- Use over-the-shoulder shots for conversations and discovery.
- Use tracking shots when characters move.
- Use push-in/pull-out for reveals and emphasis.
- Use pan/tilt to guide attention.

- Use rack-focus-like emphasis when supported by the renderer.
- Do not animate the camera continuously just for decoration; camera movement must guide the viewer.

7. CHARACTER ANIMATION SYSTEM

```
interface CharacterState {
  characterId: string;
  position: [number, number, number];
  rotation: number;
  pose: string;
  expression: string;
  gazeTarget?: string;
  action?: string;
}

interface ActionEvent {
  id: string;
  startSec: number;
  endSec: number;
  characterId?: string;
  targetId?: string;
  action: string;
  easing?: string;
  parameters?: Record<string, unknown>;
}

interface Shot {
  id: string;
  durationSec: number;
  camera: CameraPlan;
  characters: CharacterState[];
  actions: ActionEvent[];
  visualEvents: VisualEvent[];
}
```

8. CHARACTER ACTION EXECUTION

```
function executeAction(
  character: CharacterState,
  action: ActionEvent,
  t: number
) {
  const p = progress(action.startSec, action.endSec, t);

  switch (action.action) {
    case "walk":
      return walk(character, p, action.parameters);
    case "run":
      return run(character, p, action.parameters);
    case "point":
      return point(character, p, action.parameters);
    case "look_at":
      return lookAt(character, action.targetId);
    case "pick_up":
      return pickUp(character, action.targetId, p);
    case "repair":
      return repair(character, action.targetId, p);
    case "react":
      return react(character, p, action.parameters);
    case "talk":
      return talk(character, p);
    default:
      return character;
  }
}
```

9. OBJECTS MUST ALSO ANIMATE

- Machines should turn on/off.
- Doors should open/close.
- Vehicles should move.
- Tools should move with hands.
- Lights should change state.
- Broken components should visibly change when repaired.

- Graphs should update as the concept changes.
- Exploded-view parts should separate and return to assembly.
- Signals/particles should move through the system when appropriate.

10. EXPLANATION MODE — MORE THAN BOXES AND ARROWS

For each concept choose the best visual mechanism:

```

PROCESS      → sequential animation
MECHANISM    → 3D/cutaway/exploded view
FLOW        → particles/arrows/signals moving
COMPARISON   → side-by-side animated states
MATH         → Manim equation transformation
GRAPH        → progressive plotting + highlight
CIRCUIT      → animated signal/current + component interaction
PHYSICAL PART → 3D camera + component isolation
ABSTRACT IDEA → 2D motion graphics / visual metaphor

```

Cards and arrows are allowed as support,
not as the default renderer.

11. AUDIO IS THE MASTER TIMING SOURCE

```

CURRENT SCREENPLAY
  ↓
CURRENT NARRATION / DIALOGUE
  ↓
TTS
  ↓
AUDIO FILE + WORD/PHRASE TIMESTAMPS
  ↓
SHOT TIMELINE
  ↓
CHARACTER / CAMERA / OBJECT ACTIONS
  ↓
CAPTIONS
  ↓
FINAL RENDER

```

Never reuse previous audio.

Audio asset must include:

```

projectVersion
sceneHash
scriptHash
voiceId
voiceSettings
duration
transcriptHash

```

If any of these change, invalidate the audio.

12. AUDIO-DRIVEN ACTION WINDOWS

Example:

Transcript:

"The motor starts rotating."

Timestamp:

12.40s → 14.10s

Action plan:

```

12.40s camera moves toward motor
12.65s power indicator activates
12.85s rotor begins acceleration
13.20s rotor reaches normal speed
13.60s subtle vibration appears
13.90s character reacts
14.10s camera holds

```

This makes the narration and video feel like one production.

13. TRANSITION FROM ONE SHOT TO THE NEXT

- Use motivated transitions: camera follows movement, object close-up, whip/pan, match cut, dissolve, environmental transition.
- Do not transition every shot with the same generic fade.
- Preserve spatial continuity where the viewer needs it.
- Use establishing shots after a major location change.
- Keep important objects in understandable screen positions.

14. AI DIRECTOR PROMPT

You are the Animation Director.

Do not make a slideshow.
Do not use static characters to represent active events.

Read the approved screenplay and ask:
"What must the viewer SEE happening at this exact moment?"

For every sentence:

1. identify the event
2. identify who performs it
3. identify what object/environment changes
4. choose camera framing
5. define character action
6. define expression/gaze
7. define object interaction
8. define environment motion
9. assign exact timing from audio
10. verify that the visual event represents the spoken content

If the narration says:
"the villagers ran toward the gate",
the video must show villagers moving toward the gate.

If the narration says:
"the monster emerged from the forest",
show the emergence.

If the narration says:
"Eira warns Arin",
show Eira addressing Arin and Arin reacting.

Never replace a meaningful action with a text card.

15. VISUAL COVERAGE SCORING

For each narration segment calculate:

$$\text{coverage} = \frac{\text{meaningful_visual_events}}{\text{important_narrative_events}}$$

Target:

- >= 0.90 PASS
- 0.75–0.89 WARNING
- < 0.75 FAIL

Also calculate:
 $\text{actionDensity} = \text{meaningfulEvents} / \text{sceneDurationSeconds}$

Target for normal story scenes:
>= 0.20 meaningful events/sec

Adjust thresholds for intentional pauses,
emotional holds and establishing shots.

16. AUTOMATIC QA

- Flag any shot with no character/object/camera/environment change for >2–3 seconds unless explicitly marked as intentional hold.
- Flag narration sentences with no mapped visual event.
- Flag visual events that are not connected to any screenplay/narration event.
- Flag a character who is supposed to walk/run but has no movement event.

- Flag dialogue shots where the speaker does not animate speaking or address the listener.
- Flag old audio whose hash does not match the current screenplay.
- Flag missing character/environment assets.
- Flag broken technical text and unsupported glyphs.
- Flag caption timestamps that do not match current audio.
- Block final render on critical failures.

17. REGENERATION BEHAVIOR

User clicks:
"Regenerate Shot 06"

System:
1. keep project version
2. keep approved story
3. keep character identities
4. keep current audio unless narration changed
5. regenerate only Shot 06 visual plan/assets
6. run continuity QA
7. run audio/visual mapping QA
8. replace Shot 06
9. re-render final composition

If narration changed:
→ invalidate Shot 06 audio
→ regenerate TTS
→ re-time Shot 06
→ update captions
→ render again.

18. IMPLEMENTATION PRIORITY

Priority	Feature	Reason
P0	Canonical screenplay + shot schema	Everything else depends on this.
P0	Audio/script hashing	Prevents stale audio.
P0	Narration → visual-event mapping	Fixes story/video mismatch.
P1	Character action engine	Walking, looking, talking, interacting, reacting.
P1	Camera director	Wide/medium/close-up/tracking/push/pan.
P1	Object/environment animation	Doors, machines, lights, particles, vehicles, etc.
P1	Action-density QA	Stops static scenes.
P2	Advanced 3D/exploded/cutaway	Adds professional explanation visuals.
P2	Character reference/continuity	Improves long-form story quality.
P3	Advanced cinematic effects	Polish after core correctness.

19. MASTER IMPLEMENTATION PROMPT — SEND TO THE CODING AI

I have uploaded the V4 Animation Upgrade specification.

Implement this into my EXISTING AI educational video generator.
Do not only explain the changes. Modify the actual codebase.

IMPORTANT:

The current generated video is already visually decent, but it still contains too much static staging. The next goal is MORE REAL ANIMATION AND ACTION.

Do NOT simply add more text, cards, arrows, zooms or fades.

Implement:

1. canonical screenplay/shot schema
2. sentence-to-visual-event mapping
3. audio-driven timing
4. persistent character identities
5. character movement/actions
6. facial expression and gaze changes
7. character-to-character interaction
8. object interaction
9. environment animation
10. cinematic camera movement
11. explanation-mode technical animation
12. action-density QA
13. audio hash/version protection
14. captions ON/OFF
15. shot-level regeneration

For STORY MODE:

A story beat must become multiple shots.

A shot must contain actual actions.

Example:

"Monsters emerged from the forest."

DO NOT:

show two standing characters + paragraph.

DO:

forest movement
→ monster silhouette
→ monster steps out
→ second monster follows
→ villager notices
→ villager turns
→ facial reaction
→ camera changes
→ villagers begin moving

For every important narration sentence ask:

"What is physically happening right now?"

Then animate that event.

For EXPLANATION MODE:

Use:

2D animation
3D models
exploded views
cutaways
graphs
equations
signals
particles
simulations
camera movement
progressive diagrams
real object/component transformations

Do not use boxes/arrows/cards as the primary visual mechanism.

AUDIO:

Never reuse previous audio.

Audio validity must depend on:

projectVersion + sceneHash + scriptHash + voiceId + voiceSettings.

If narration changes, invalidate and regenerate the affected audio.

Use timestamps to drive visual actions.

QA:

Reject scenes with long unexplained static periods.

Reject important narration with no visual event.

Reject mismatched screenplay/audio.
Reject broken text.
Reject character continuity failures.

Do not replace existing working features unnecessarily.
Inspect the current project first, then implement the minimum architectural changes required.

FIRST TEST:

Create a Story Mode demo:
"A student discovers why a motor stopped working."

It must visibly show:
student enters workshop
→ notices motor
→ approaches motor
→ inspects it
→ opens/reveals component
→ second character joins
→ dialogue
→ diagnosis
→ repair
→ motor activates
→ characters react

Use multiple camera shots and meaningful action throughout.

SECOND TEST:

Create an Explanation Mode demo:
"How a DC motor works."

It must show actual motor parts, an exploded/internal view,
rotor movement, magnetic/field explanation and synchronized narration.

Do not call the project complete until these tests visibly demonstrate
real actions instead of slideshow-like staging.

At the end report:

- files changed
- new architecture
- action system
- audio invalidation system
- synchronization system
- QA results
- test video results
- remaining limitations

20. DEFINITION OF DONE

The upgrade is complete only when the viewer can watch the generated video and clearly see events happening rather than being told about events while static graphics remain on screen.

Success test: mute the audio and watch the video. A viewer should still be able to understand the major sequence of actions from the visuals alone. Then listen to the audio: the narration should describe the same events at the same time.

Final target: the software should behave like an AI animation director—not an AI PowerPoint generator.